Q. What is the use of search() in Regular Expression? Explain with an example.

Answer:

In Python, the search() function is used to search for a pattern in a string. It searches the entire string and returns the first match found. If no match is found, it returns None.

Syntax:

re.search(pattern, string)

Example:

```
import re

text = "My name is Apurva and I am learning Python"

result = re.search("Python", text)

if result:

   print("Pattern found")

else:

   print("Pattern not found")
```

Output:

Pattern found

Explanation:

- import re – Imports the regular expression module.
- re.search() – Searches for the given pattern in the string.
- "Python" – The pattern we want to find.
- text – The string in which we are searching.
- If the pattern is found, it returns a match object; otherwise, it returns None.

Remember the difference:

Function	**Use**
re.search()	Finds the first match anywhere in the string.
re.match()	Checks for a match only at the beginning of the string.
re.findall()	Returns all matching results in a list.

Easy way to remember: search() looks throughout the entire string and returns only the first match.